

V5 Real-World External Battery Report

Run label: realworld_battery_40datasets_full_run

Loaded dataset count recorded in registry: 228

Dataset-level primary decision rows: 809

Primary strong-win rows before dataset-level BH-FDR: 256

Rows surviving BH-FDR at 5%: 255

Rows surviving BH-FDR at 10% sensitivity: 257

Saturated datasets flagged by the 10-win cap: 21

Secondary oracle audit rows: 5900

Interpretation principle: the primary claim is not the raw number of gate-row wins. The primary claim is the number, profile, and concentration of eligible in-regime dataset-level signals after paired CI and BH-FDR correction.

Secondary depth probes and oracle rows are transparency layers, not independent discoveries.

No Free Lunch interpretation: broad failure to dominate is not a weakness. It is evidence that the GA specialist is conditional. The important signal is whether the few surviving gains are concentrated in the predicted empirical profiles rather than scattered across unrelated regimes.

Pre-registered winner taxonomy

This page defines the final tournament grid used in the CSVs and PDF.

1. Eligibility layer: eligible means the empirical dataset profile matches the specialist family under fixed thresholds. Not eligible means control or cross-regime audit, not a primary loss or win.
2. Regime layer: in_regime rows define the confirmatory path. cross_regime_audit rows are retained for transparency and to detect accidental wins outside the predicted regime.
3. Benchmark layer: primary_preregistered uses the fixed benchmark assigned by profile. Oracle comparisons are written separately and must not be promoted to the primary claim.
4. Evidence layer: strong_win requires joint improvement and CI support; strong_point_signal has positive gate evidence but not full CI support; pareto_signal improves one metric without material harm to the other; equal_weight_gain shows learned weights beat naive equal weights; no_win means the benchmark remains dominant.
5. CI layer: ci_confirmed requires the paired bootstrap interval for the reported gain to stay above zero. near_ci means the point estimate is positive but the interval touches zero. descriptive means no statistical support.
6. FDR layer: dataset_level_q_value_bh is the primary multiplicity correction. Gate-row FDR is retained for audit only because condition rows within a dataset are correlated.

Audit policies embedded in the output CSVs

These policies are also written into `external_methods_audit_flags.csv` and relevant decision tables.

Profile gate policy: Pre-registered empirical profile gate: bounded/proportion by hint/domain; zero-inflated/count by hint/domain; normal_like when $\text{abs}(\text{skew}) \leq 0.50$ and $\text{excess_kurtosis} \leq 1.00$; heavy_tail_symmetric when $\text{abs}(\text{skew}) \leq 1.00$ and $\text{excess_kurtosis} > 3.00$; lognormal_like when $\text{skew} > 1.00$ with positive support; weibull_like when $\text{skew} > 0.50$ with positive support; otherwise empirical_unknown.

FDR policy: Benjamini-Hochberg FDR is computed at dataset level on `external_dataset_level_decisions.csv` for primary eligible in-regime rows only. Primary $\alpha=0.05$; secondary sensitivity $\alpha=0.1$.

Output layout: `01_results_basic` contains compact final results; `02_reports` contains generated report files; `03_audit_ready` contains full provenance, decisions, oracle audit, and split CSVs; `98_discarded_nonfinal` is reserved for obsolete or nonfinal byproducts; `99_recovery_logs` contains downloads, checkpoints, temporary recovery outputs, and logs.

CI policy: paired bootstrap aligned to the same reported gain statistic

Saturation cap: 10 countable primary wins per dataset.

Depth probes secondary only: TRUE.

Dataset-level coverage by empirical profile

A balanced external battery should avoid concentration in a single profile or catalog source.

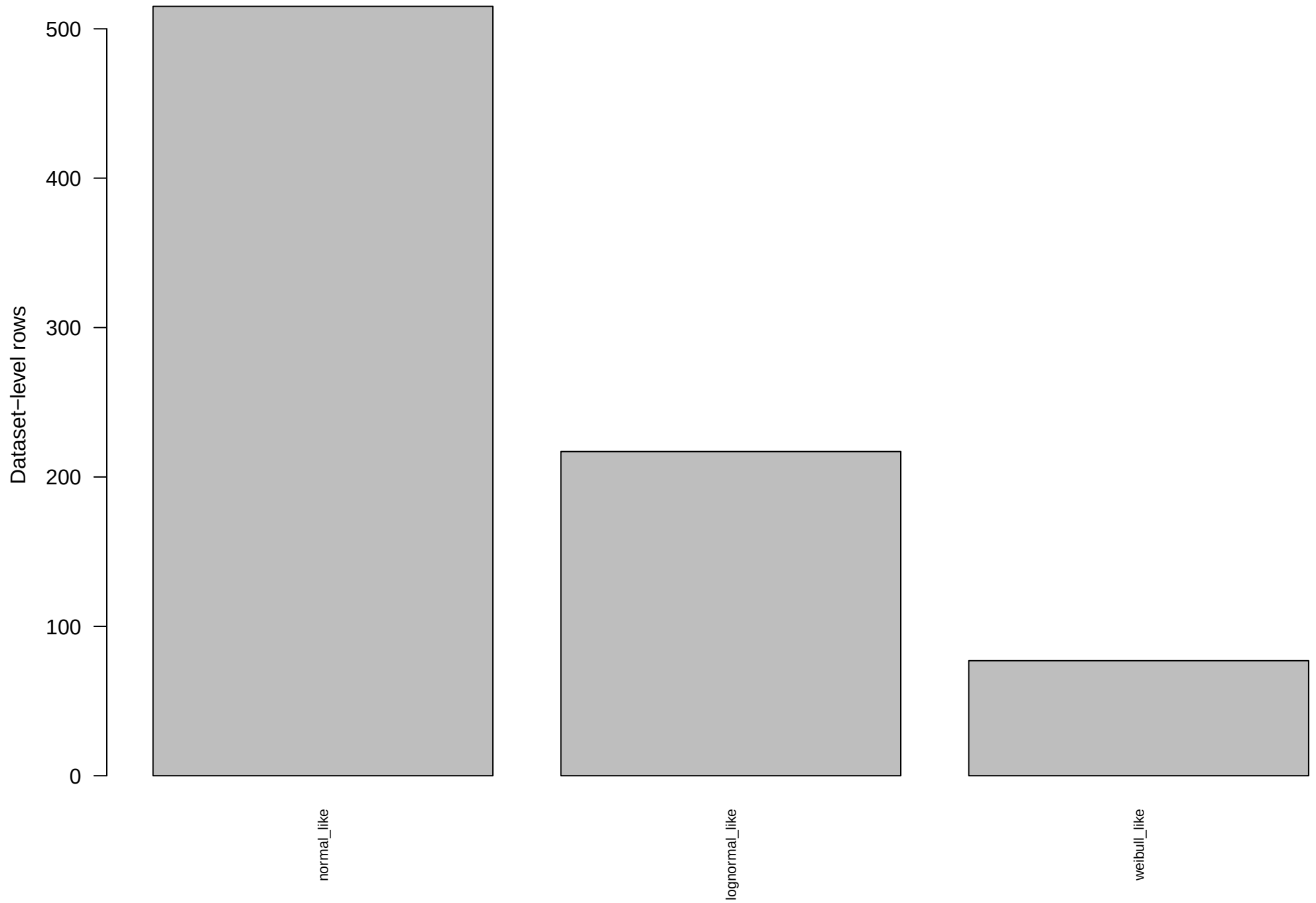


Figure legend: counts are computed from V5 final CSVs. Primary interpretation should use dataset-level FDR columns, not raw gate-row totals.

Best evidence layer by dataset-level decision

This is the main hierarchy of evidence. Strong wins are the only primary winners; Pareto and equal-weight outcomes are supporting evidence.

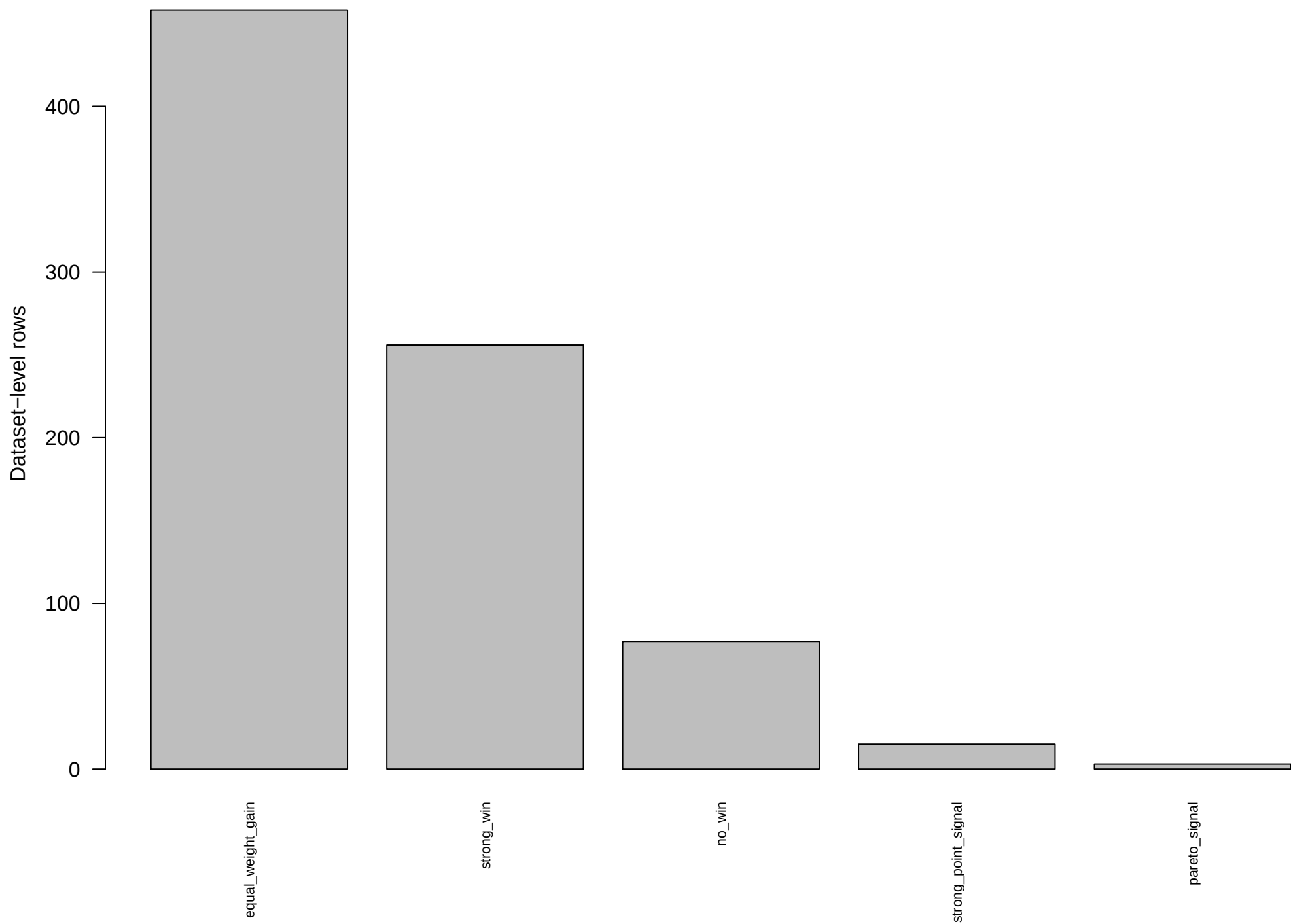


Figure legend: counts are computed from V5 final CSVs. Primary interpretation should use dataset-level FDR columns, not raw gate-row totals.

CI layer distribution

The paired bootstrap interval is aligned to the same gain statistic reported in the decision table.

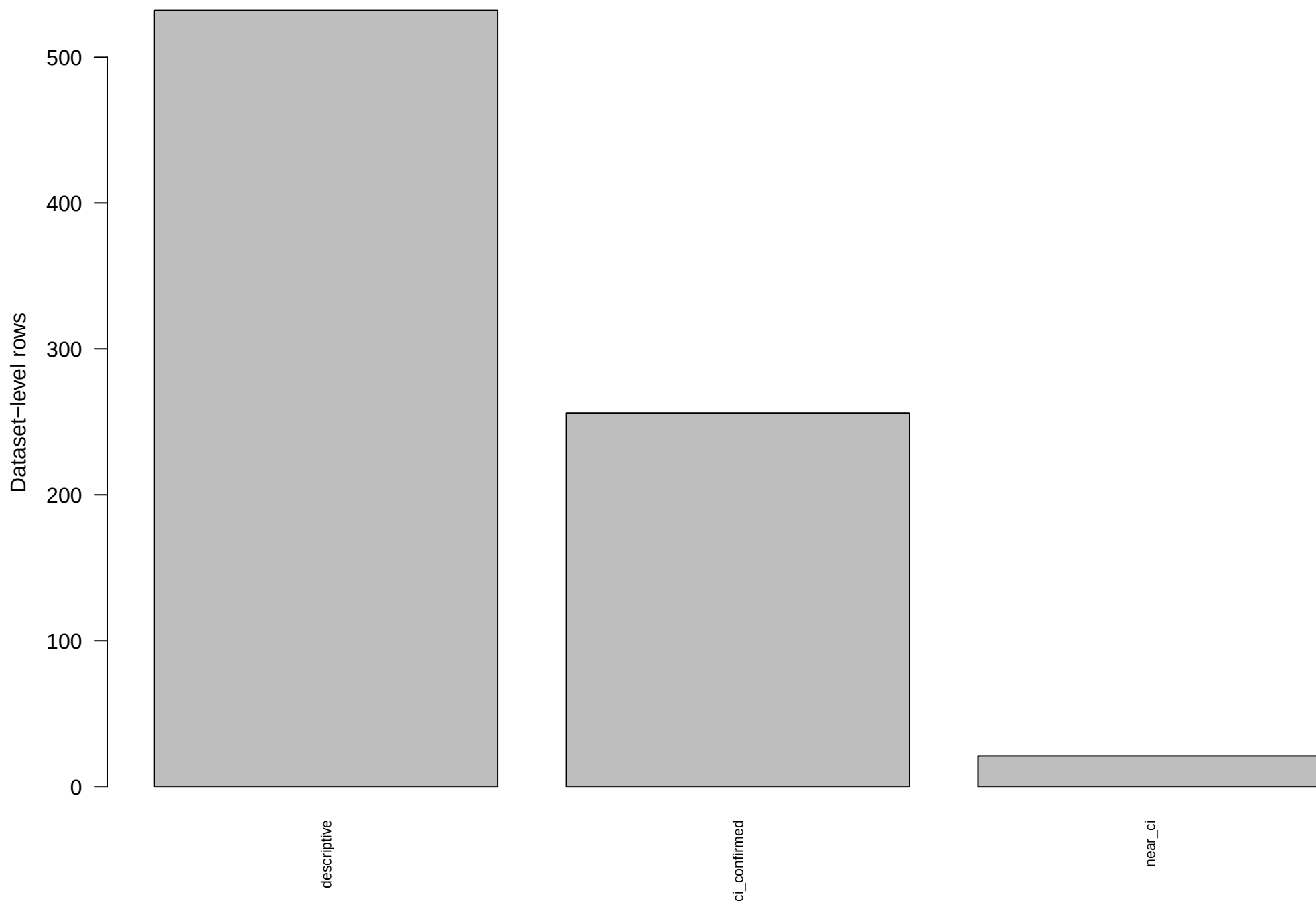


Figure legend: counts are computed from V5 final CSVs. Primary interpretation should use dataset-level FDR columns, not raw gate-row totals.

Dataset-level BH q-value distribution

The 5% and 10% FDR flags in the CSVs should be used as the final multiplicity layer.

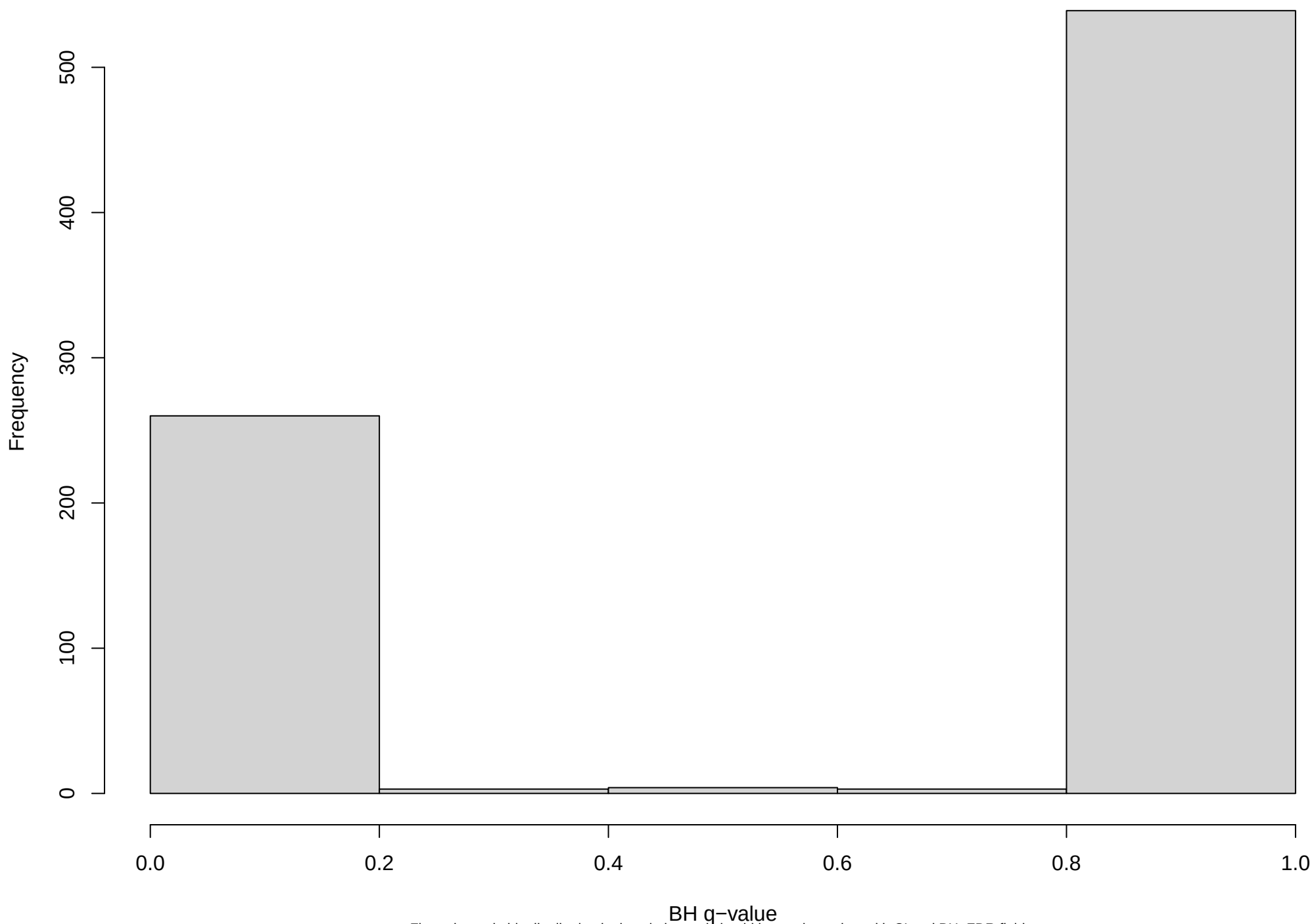


Figure legend: this distribution is descriptive and should be read together with CI and BH-FDR fields.

Top dataset–level q95 gains

Sorted by best q95 gain. This table is descriptive until CI and FDR columns are checked.

dataset_id	empirical_profile	specialist_family	best_gain_q95	best_ci_low	dataset_level_q_va	best_evidence_laye
UCI_online_news_tokens__dept	lognormal_like	lognormal	31287.0384811903	29975.8529718265	0	strong_win
UCI_online_news_tokens	lognormal_like	lognormal	12854.7633107162	12234.0676318599	0	strong_win
UCI_online_news_tokens__dept	lognormal_like	lognormal	12359.7443471632	11878.4563495539	0	strong_win
UCI_online_news_tokens__dept	lognormal_like	lognormal	9326.8907254329	8512.83839282266	0	strong_win
UCI_online_news_tokens__dept	lognormal_like	lognormal	8271.33922581111	7722.7731236642	0	strong_win
RD_ISLR_Default_auto	normal_like	normal	7554.69883276416	6954.21863768991	0	strong_win
UCI_bike_hour_count__depth_u	lognormal_like	lognormal	7421.02780945269	7273.65364787284	0	strong_win
RD_ISLR_Default_auto__depth_	normal_like	normal	7206.27059703691	6838.96605378919	0	strong_win
RD_ISLR_Default_auto__depth_	normal_like	normal	6811.08711259654	6011.66165369961	0	strong_win
UCI_online_news_tokens__dept	lognormal_like	lognormal	5370.30818692483	5212.44344152146	0	strong_win
RD_bayesrules_bird_counts_au	lognormal_like	lognormal	5085.90979594456	4674.12392231865	0	strong_win
RD_ISLR_Default_auto__depth_	normal_like	normal	4918.35360386581	4685.19489377988	0	strong_win
UCI_bike_hour_registered__de	lognormal_like	lognormal	4891.58013606806	4784.02146379585	0	strong_win
UCI_bike_hour_count	lognormal_like	lognormal	4888.7112520839	4771.69892706106	0	strong_win
UCI_bike_hour_count__depth_b	lognormal_like	lognormal	4885.76349428187	4779.44951053438	0	strong_win

Showing 15 of 15 rows.

Rows surviving dataset-level BH-FDR at the sensitivity threshold

These are the most defensible candidates. Use the 5% flag for the strict claim and 10% as sensitivity analysis.

dataset_id	empirical_profile	specialist_family	best_gain_q95	best_ci_low	dataset_level_q_va	best_final_evidenc
RD_admiral_admiral_adlb_auto	weibull_like	weibull	0.000229740185762918	0.000228964950952431	0	eligible / in_regime / prima
RD_admiral_admiral_adlb_auto	weibull_like	weibull	0.000262702830716009	0.00025704551944726	0	eligible / in_regime / prima
RD_admiral_admiral_adlb_auto	weibull_like	weibull	0.000389697424269722	0.000387087746939702	0	eligible / in_regime / prima
RD_admiral_admiral_adlb_auto	weibull_like	weibull	0.000388348948901069	0.000386154750777342	0	eligible / in_regime / prima
RD_admiral_admiral_adlb_auto	weibull_like	weibull	0.000376606989611476	0.0003733410975858	0	eligible / in_regime / prima
RD_admiral_admiral_adlb_auto	weibull_like	weibull	0.000375002526357883	0.000373912004553934	0	eligible / in_regime / prima
RD_admiral_admiral_adlb_auto	lognormal_like	lognormal	0.00120012963655923	0.00115580051090111	0	eligible / in_regime / prima
RD_admiral_admiral_adlb_auto	lognormal_like	lognormal	0.00135833248740615	0.00120042008906457	0	eligible / in_regime / prima
RD_admiral_admiral_adlb_auto	lognormal_like	lognormal	0.00693935121989858	0.00674304670052542	0	eligible / in_regime / prima
RD_admiral_admiral_adlb_auto	lognormal_like	lognormal	0.00682901762615354	0.00665519324969508	0	eligible / in_regime / prima
RD_alone_loadouts_auto	normal_like	normal	0.000817062594063594	0.000369221609729723	0	eligible / in_regime / prima
RD_alone_loadouts_auto	normal_like	normal	0.000434085328362531	-3.25201192226803e-05	0.098632944228275	eligible / in_regime / prima
RD_alone_loadouts_auto	normal_like	normal	0.0011757369247322	0.000765931283970743	0	eligible / in_regime / prima
RD_alone_loadouts_auto	normal_like	normal	0.00152243461842912	0.000941910902942053	0	eligible / in_regime / prima
RD_archdata_Handaxes_auto__d	weibull_like	weibull	0.581494487009282	0.415498041009523	0	eligible / in_regime / prima

Showing 15 of 15 rows.

Saturated datasets

Saturation prevents one dataset from inflating the primary N. Additional signal remains visible but not count-inflating.

dataset_id	empirical_profile	specialist_family	n_countable_primar	n_uncapped_primary	dataset_saturated	dataset_stop_reaso
RD_archdata_Handaxes_auto	lognormal_like	lognormal	7	7	TRUE	
RD_asaur_prostateSurvival_au	weibull_like	weibull	7	7	TRUE	
RD_bakeoff_challenges_data__	normal_like	normal	10	10	TRUE	
RD_bayesrules_bird_counts_au	lognormal_like	lognormal	1	1	TRUE	
RD_boot_amis_auto	normal_like	normal	10	10	TRUE	
RD_camerondata_vietnam_ind_a	lognormal_like	lognormal	7	7	TRUE	
RD_dragracer_rpdr_contep_aut	normal_like	normal	10	10	TRUE	
RD_fpp2_calls_auto	normal_like	normal	10	10	TRUE	
RD_gt_pizzaplace_auto	normal_like	normal	10	10	TRUE	
RD_HistData_Pollen_auto	normal_like	normal	10	10	TRUE	
RD_ISLR_Default_auto	normal_like	normal	10	10	TRUE	
RD_OncoDataSets_ProstateSurv	weibull_like	weibull	7	7	TRUE	

Showing 12 of 12 rows.

Taxonomy grid counts

This grid cross-tabulates eligibility, regime, benchmark, evidence, and CI layers.

eligibility_layer	regime_layer	benchmark_layer	evidence_layer	ci_layer	Freq
eligible	in_regime	primary_preregistered	equal_weight_gain	ci_confirmed	0
eligible	in_regime	primary_preregistered	no_win	ci_confirmed	0
eligible	in_regime	primary_preregistered	pareto_signal	ci_confirmed	0
eligible	in_regime	primary_preregistered	strong_point_signal	ci_confirmed	0
eligible	in_regime	primary_preregistered	strong_win	ci_confirmed	2211
eligible	in_regime	primary_preregistered	equal_weight_gain	descriptive	2774
eligible	in_regime	primary_preregistered	no_win	descriptive	778
eligible	in_regime	primary_preregistered	pareto_signal	descriptive	7
eligible	in_regime	primary_preregistered	strong_point_signal	descriptive	0
eligible	in_regime	primary_preregistered	strong_win	descriptive	0
eligible	in_regime	primary_preregistered	equal_weight_gain	near_ci	26
eligible	in_regime	primary_preregistered	no_win	near_ci	4
eligible	in_regime	primary_preregistered	pareto_signal	near_ci	8
eligible	in_regime	primary_preregistered	strong_point_signal	near_ci	92
eligible	in_regime	primary_preregistered	strong_win	near_ci	0

Showing 15 of 15 rows.

Specialist–domain survival map

This table identifies where specialist families retain useful transfer signal and where benchmark control dominates.

specialist_id	domain	n_rows	n_strong_win	n_pareto_signal	n_equal_weight_gai	median_q95_gain	best_dataset
TAX-Q1R008	biological_positive	36	24	0	9	0.00403432041813757	UCI_abalone_whole_weight
TAX-Q1R008	biomedical_signal	9	0	0	9	-1.90988329702594e-05	UCI_parkinsons_PPE
TAX-Q1R008	engineering_positive	9	0	0	9	-26.8842375683425	UCI_auto_horsepower
TAX-Q1R008	environmental_extremes	9	0	0	5	-0.0971092325966527	UCI_forestfires_log_area
TAX-Q1R008	mobility_demand	108	94	0	5	1023.0666695088	UCI_bike_hour_count__depth_u
TAX-Q1R008	positive_skew_tabular	144	78	0	48	0.0077935500405232	UCI_wine_white_residual_suga
TAX-Q1R008	rdatasets_admiral	27	15	0	0	0.00120636338456362	RD_admiral_admiral_adlb_auto
TAX-Q1R008	rdatasets_archdata	9	3	0	6	-1.38907421045384	RD_archdata_Handaxes_auto
TAX-Q1R008	rdatasets_bayesrules	36	30	0	0	1719.58194522305	RD_bayesrules_bird_counts_au
TAX-Q1R008	rdatasets_camerondata	27	3	0	24	-0.0273171588128601	RD_camerondata_vietnam_ind_a
TAX-Q1R008	rdatasets_collegeScorecard	9	0	0	0	-939816.041305793	RD_collegeScorecard_scorecar
TAX-Q1R008	rdatasets_crch	9	0	0	6	-2.68290760648687	RD_crch_Rainlbk_auto
TAX-Q1R008	rdatasets_DAAG	9	0	0	9	-30373.3363582844	RD_DAAG_nassCDS_auto
TAX-Q1R008	web_popularity	72	66	0	6	0.0856010710047976	UCI_online_news_tokens__dept
TAX-Q1R009	biological_positive	118	64	0	39	0.00136182751270069	UCI_abalone_whole_weight_de
TAX-Q1R009	biomedical_signal	39	0	0	39	-1.88408536191688e-05	UCI_parkinsons_PPE
TAX-Q1R009	engineering_positive	39	0	0	38	-22.0499379421892	UCI_auto_horsepower
TAX-Q1R009	environmental_extremes	39	0	0	26	-0.0864673501536853	UCI_forestfires_log_area

Showing 18 of 18 rows.

Substantive interpretation

Suggested reading of V5 outputs.

The strict primary result is the FDR-adjusted dataset-level count: 255 rows at 5% and 257 rows at 10% sensitivity.

This should be reported before any larger condition-level or depth-probe counts.

If surviving rows concentrate in lognormal_like or weibull_like profiles, the correct interpretation is not universal superiority. It is regime-conditional transfer: GA specialists retain value where the external dataset resembles the discovery regime.

If most datasets remain no_win or near_ci, that supports the No Free Lunch frame: robust benchmarks continue to dominate outside the specialist regime, and gains are localized rather than universal.

If a dataset becomes saturated, the cap protects the paper. It says the dataset contains strong repeated signal, but the primary count is limited so the result cannot be driven by one public source or one data structure.

Depth probes should be discussed as within-dataset mechanism checks. They can deepen the explanation of a signal but should never be counted as independent datasets.

The oracle audit answers a reviewer question: what happens against the best available benchmark? It is intentionally secondary because an oracle benchmark is stronger than a pre-registered real comparator and can be biased by selection across many benchmarks.